Lecture 4: Using Basic Integration Formula

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Perform basic integration by using basic integration formula.
- ☐ Apply suitable substitution in computing integrals.

Introduction

Recall that if $F(x)$ is a differentiable function of x with $F(x) = f'(x)$, then

$$\int f(x)\,dx = F(x) + C.$$

Fundamental Theorem of Calculus.

Also recall that if we let $u = g(x)$, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Substitution method.

Eg: $\frac{d}{dx}\tanh^{-1}u = \frac{1}{1-u^2}$, $|u|<1$

$$\int \frac{1}{1-u^2}\,du = \tanh^{-1}u + C$$

Let $u = \frac{x}{a}$ ($g(x)$) $\Rightarrow$ $\int \frac{1}{1-\left(\frac{x}{a}\right)^2}\frac{1}{a}\,dx = \tanh^{-1}\left(\frac{x}{a}\right) + C$

$du = \frac{1}{a}\,dx$

$\Rightarrow \int \frac{1}{a^2 - x^2}\,dx = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C$

Basic Integration Formulas

1. $\int k\,dx = kx + C$ (any number k)

2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

3. $\int \frac{dx}{x} = \ln|x| + C$

4. $\int e^x\,dx = e^x + C$

5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$

6. $\int \sin x\,dx = -\cos x + C$

7. $\int \cos x\,dx = \sin x + C$

8. $\int \sec^2 x\,dx = \tan x + C$

9. $\int \csc^2 x\,dx = -\cot x + C$

10. $\int \sec x \tan x\,dx = \sec x + C$

11. $\int \csc x \cot x\,dx = -\csc x + C$

12. $\int \tan x\,dx = \ln|\sec x| + C$

13. $\int \cot x\,dx = \ln|\sin x| + C$

14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$

16. $\int \sinh x\,dx = \cosh x + C$

17. $\int \cosh x\,dx = \sinh x + C$

18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$

19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$

20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\operatorname{arcsec}\left|\frac{x}{a}\right| + C$

21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C \quad (a > 0)$

22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C \quad (x > a > 0)$

23. $\int \frac{dx}{x\sqrt{a^2 - x^2}} = -\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{x}{a}\right) + C$

for $\int \frac{dx}{x^2 - a^2}$ or $\int \frac{dx}{a^2 - x^2}$, you will learn to do it using partial fraction.

Examples

$$\int_3^5 \frac{2x-3}{\sqrt{x^2-3x+1}}\,dx$$

diff. ⇓

$2x-3$

Let $u = x^2-3x+1$

$du = 2x-3\,dx$

$x=3,\ u=1$

$x=5,\ u=11$

$$= \int_1^{11} \frac{du}{\sqrt{u}}$$

$$= \left[2\sqrt{u}\right]_1^{11}$$

$$= 2\sqrt{11} - 2\sqrt{1} = 2\sqrt{11} - 2$$

$$\int \frac{dx}{\sqrt{8x-x^2}} = \int \frac{dx}{\sqrt{-(x^2-8x)}}$$

$$= \int \frac{dx}{\sqrt{-[(x-4)^2-16]}}$$

$$= \int \frac{dx}{\sqrt{16-(x-4)^2}}$$

Let $u = x-4$

$du = dx$

$$= \int \frac{du}{\sqrt{(4)^2-u^2}} = \arcsin\left(\frac{u}{4}\right)+C$$

$$= \arcsin\left(\frac{x-4}{4}\right)+C$$

Examples

$$\int (\cos x \sin 2x + \sin x \cos 2x)\, dx$$

Recall: $\sin(A+B) = \sin A \cos B + \cos A \sin B$.

$$= \int \sin(x+2x)\, dx = \frac{1}{3}\int 3\sin(3x)\, dx$$

Let $u = 3x$, $du = 3dx$

$$= \frac{1}{3}\int \sin u\, du$$

$$= \frac{1}{3}(-\cos u) + C = -\frac{1}{3}\cos(3x) + C$$

$$\int_0^{\frac{\pi}{4}} \frac{dx}{1-\sin x} = \int_0^{\pi/4} \frac{1}{1-\sin x}\left(\frac{1+\sin x}{1+\sin x}\right) dx$$

$$= \int_0^{\pi/4} \frac{1+\sin x}{1-\sin^2 x}\, dx = \int_0^{\pi/4} \frac{1+\sin x}{\cos^2 x}\, dx$$

$$= \int_0^{\pi/4} \frac{1}{\cos^2 x}\, dx + \int_0^{\pi/4} \frac{\sin x}{\cos^2 x}\, dx$$

$\left(\frac{\sin x}{\cos x}\right)\left(\frac{1}{\cos x}\right)$

$$= \int_0^{\pi/4} \sec^2 x \, dx + \int_0^{\pi/4} \tan x \sec x \, dx$$

$$= \left[\tan x\right]_0^{\pi/4} + \left[\sec x\right]_0^{\pi/4}$$

$$= (\tan \pi/4 - \tan 0) + (\sec \pi/4 - \sec 0)$$

$$= (1 - 0) + (\sqrt{2} - 1) = \sqrt{2}$$

Examples

$$\int \frac{3x^2 - 7x}{3x + 2}\, dx$$

Improper fraction.

$$\begin{array}{r} x - 3 \\ 3x+2 \overline{)\, 3x^2 - 7x + 0} \\ (-)\ 3x^2 + 2x \\ \hline -9x + 0 \\ (-)\ -9x - 6 \\ \hline 6 \end{array}$$

$$= \int \underbrace{(x-3)}_{\text{quotient}} + \frac{\overbrace{6}^{\text{remainder}}}{3x+2}\, dx$$

$$= \int x - 3\, dx + \int \frac{6}{\underbrace{3x+2}_{\text{diff. } u}}\, dx \qquad \leftarrow 3 \cdot 2.$$

$$= \int x - 3\, dx + 2\int \frac{du}{u} \qquad \frac{1}{3}$$

Let $u = 3x+2$, $du = 3dx$

$$= \frac{x^2}{2} - 3x + 2\ln|u| + C$$

$$= \frac{x^2}{2} - 3x + 2\ln|3x+2| + C.$$

$$\int \frac{3x + 2}{\sqrt{1 - x^2}}\, dx$$

$$\int \frac{3x+2}{\sqrt{1-x^2}}\,dx = \int \frac{3x}{\sqrt{1-x^2}}\,dx + \int \frac{2}{\sqrt{1-x^2}}\,dx$$

diff. ⇓ $-2x$

substitution

formula no. 18.

Let $u = 1-x^2$

$du = -2x\,dx$

$$= \frac{3}{-2}\int \frac{-2x\,dx}{\sqrt{1-x^2}} + 2\int \frac{1}{\sqrt{1-x^2}}\,dx$$

$$= \frac{3}{-2}\int \frac{du}{\sqrt{u}} + 2\arcsin x$$

$$= \frac{-3}{2}\left(2\sqrt{u}\right) + 2\arcsin x + C$$

$$= -3\sqrt{1-x^2} + 2\arcsin x + C.$$

Examples

$$\int \frac{dx}{(1+\sqrt{x})^3}$$

$$= \int \frac{2\sqrt{x}}{(1+\sqrt{x})^3 \, 2\sqrt{x}} \, du \qquad 2(u-1)$$

u^3

Let $u = 1+\sqrt{x} \Rightarrow \sqrt{x} = u-1$

$du = \frac{1}{2\sqrt{x}} dx$

$d3R \cdot \Downarrow$

$\frac{1}{2\sqrt{x}} dx$

$$= \int \frac{2(u-1)}{u^3} \, du = \int \frac{2u}{u^3 u^2} \, du - \int \frac{2}{u^3} du$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^3 \cos x \, dx$$

$$= 2\int u^{-2} \, du - 2\int u^{-3} du$$

$$= 2\left(\frac{u^{-1}}{-1}\right) - 2\left(\frac{u^{-2}}{-2}\right) + C$$

$$= \frac{-2}{u} + \frac{1}{u^2} + C$$

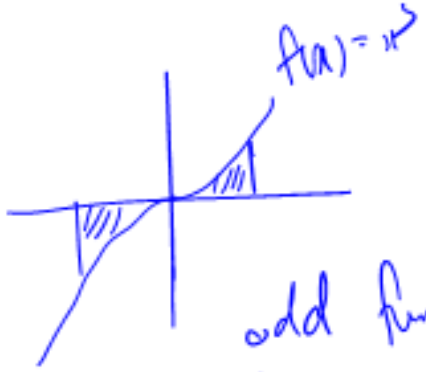
$$= \frac{1-2u}{u^2} + C$$

$$= \frac{1-2(1+\sqrt{x})}{(1+\sqrt{x})^2}$$

$$= \frac{-1-2\sqrt{x}}{(1+\sqrt{x})^2}$$

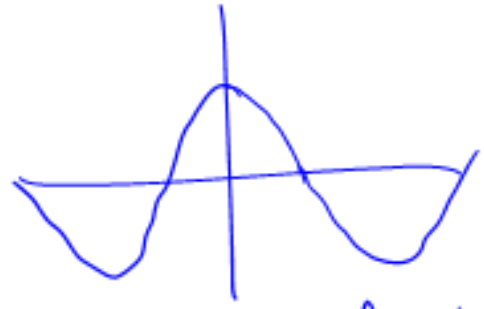
$$\int_{-\pi/2}^{\pi/2} x^3 \cos x \, dx = 0$$

Alternatively, this can be solved using integration by part. (Lec 5).

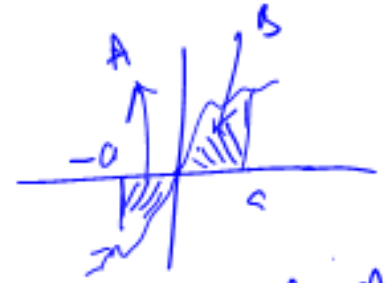


odd function.

$f(-a) = -f(a)$



even function



Area A = Area B

$x^3 \cos x \Rightarrow$ odd function.

odd function × even function

Let $h(x) = x^3 \cos x$,

$$h(-x) = \underline{(-x)^3} \ \underline{\cos(-x)}$$

$$= -x^3 \cos x = -h(x)$$

(odd function)

$$\int_{-\pi/2}^{0} h(x)\,dx = -\int_{0}^{\pi/2} h(x)\,dx$$

$$\Rightarrow \int_{-\pi/2}^{0} h(x)\,dx + \int_{0}^{\pi/2} h(x)\,dx = 0$$

Examples

Evaluate $\int_0^1 \frac{x^3}{(4+x^2)^{\frac{5}{2}}}\,dx$ by using the substitution

- $x = 2\tan\theta$

You will see more on this in Lecture 7.

- $x = 2\sinh\theta$

Exercise (very similar to $x = 2\tan\theta$)

Let $I = \int_0^1 \frac{x^3}{(4+x^2)^{\frac{5}{2}}}\,dx$

$x = 0,\ 2\tan\theta = 0 \Rightarrow \theta = 0$

$x = 1,\ 2\tan\theta = 1 \Rightarrow \theta = \arctan(\frac{1}{2})$

Let $x = 2\tan\theta$.

$dx = 2\sec^2\theta\,d\theta$

$$I = \int_0^{\arctan(\frac{1}{2})} \frac{(8\tan^3\theta)(2\sec^2\theta\,d\theta)}{(4+4\tan^2\theta)^{\frac{5}{2}}}$$

x^3, dx, $(4+x^2)^{5/2}$

$$= \int_0^{\arctan(\frac{1}{2})} \frac{16\tan^3\theta\sec^2\theta\,d\theta}{[4(1+\tan^2\theta)]^{\frac{5}{2}}}$$

$1+\tan^2\theta = \sec^2\theta$

$$= \int_0^{\arctan(\frac{1}{2})} \frac{16\tan^3\theta\sec^2\theta\,d\theta}{(4\sec^2\theta)^{\frac{5}{2}}}$$

$$= \int_0^{\arctan(1/2)} \frac{16 \tan^3\theta \sec^2\theta \, d\theta}{\left(\sqrt{4\sec^2\theta}\right)^5}$$

$$= \int_0^{\arctan(1/2)} \frac{16 \tan^3\theta \sec^2\theta \, d\theta}{(2\sec\theta)^5}$$

$$= \int_0^{\arctan(1/2)} \frac{16 \tan^3\theta \sec^2\theta}{32 \sec^5\theta} \, d\theta$$

$$= \frac{1}{2} \int_0^{\arctan(1/2)} \frac{\tan^3\theta}{\sec^3\theta} \, d\theta$$

$$= \frac{1}{2} \int_0^{\arctan(1/2)} \frac{\sin^3\theta}{\cos^3\theta} \left(\cos^3\theta\right) d\theta$$

$$= \frac{1}{2}\int_0^{\arctan(1/2)} \sin^3\theta \, d\theta$$

$$= \frac{1}{2}\int_0^{\arctan(1/2)} \sin\theta(\sin^2\theta)\, d\theta$$ Lecture 6.

$$= -\frac{1}{2}\int_0^{\arctan(1/2)} -\sin\theta\,(1-\cos^2\theta)\, d\theta$$ du

Let $u = \cos\theta$

$du = -\sin\theta\, d\theta$

$\theta = 0,\ u = 1$

$\theta = \arctan(1/2),\ u = \frac{1}{\sqrt{5}}$

$$= -\tfrac{1}{2}\int_1^{1/\sqrt{5}} (1-u^2)\, du$$

$$= -\tfrac{1}{2}\left[u - \frac{u^3}{3}\right]_1^{1/\sqrt{5}}$$

$$\approx 0.00537$$



Examples

Evaluate $\int_0^1 \frac{x^3}{(4+x^2)^{\frac{5}{2}}}\,dx$ by using the substitution

Let $I = \int_0^1 \frac{x^3}{(4+x^2)^{\frac{5}{2}}}\,dx$

Let

- $u^2 = 4 + x^2 \Rightarrow u = \sqrt{4+x^2}$

implicit differentiation

$\Rightarrow x^2 = u^2 - 4$

$2u\,du = 2x\,dx$

$u\,du = x\,dx$

$du = \frac{x}{u}\,dx = \frac{x}{\sqrt{4+x^2}}\,dx$

$I = \int_0^1 \frac{x^2 \cdot x\,dx}{(\sqrt{4+x^2})^4 \cdot \sqrt{4+x^2}}$ (du)

u^4

$= \int_2^{\sqrt{5}} \frac{u^2-4}{u^4}\,du$

- $u = 4 + x^2$

A "usual" substitution

(Exercise)

$x = 0 \Rightarrow u^2 = 4 \Rightarrow u = 2$

$x = 1 \Rightarrow u^2 = 5 \Rightarrow u = \sqrt{5}$

$= \int_2^{\sqrt{5}} u^{-2} - 4u^{-4}\,du$

$= \left[\frac{u^{-1}}{-1}\right]_2^{\sqrt{5}} - \left[\frac{4u^{-3}}{-3}\right]_2^{\sqrt{5}}$

≈ 0.00537617